

On asymptotic normality of cross data matrix-based PCA in high dimension low sample size

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ABSTRACT

Principal component analysis in high dimension low sample size setting has been an active research area in recent years. Yata and Aoshima (2010) proposed a cross data matrix-based method and showed the asymptotic normality for estimates of spiked eigenvalues and also consistency for corresponding estimates of PC directions. However, the asymptotic normality for estimates of PC directions is still lacking. In this article, we have extended Yata and Aoshima (2010)'s work to include the investigation of the asymptotic normality for the leading CDM-based PC directions and to compare it with the asymptotic normality for the classical PCA. Numerical examples are provided to illustrate the asymptotic normality.

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1. Introduction

The study of principal component analysis (PCA) has a long history in statistics and many other scientific fields. Early works of asymptotic theory can be found in Anderson [2] and Tyler [13] in the setting of large sample size and fixed dimension. Later, Bai [5], Baik et al. [6], Paul [11] and many others have investigated the asymptotic behavior of empirical eigenvalues and eigenvectors in the setting of both sample size and dimension diverging. Recently, great attention has been given to the high dimension and low sample size (HDLSS) data due to its wide applications such as genetic microarrays, medical images, text recognition, finance, chemometrics, etc. See, for instance, Johnstone and Lu [9], Hall et al. [7], Ahn et al. [1], Yata and Aoshima [15,16] and Shen et al. [12].

Yata and Aoshima [16] proposed a cross data matrix (CDM) method for PCA and demonstrated its superior empirical performance than the classical PCA in HDLSS setting. In a study of prostate cancer, they showed that the first two PC directions of CDM-PCA can better separate the normal from the tumor samples than PCA. Yata and Aoshima [17] proposed another effective high-dimensional PCA method, the noise reduction (NR) method, and gave asymptotic properties of the NR method of eigenvalues and PC directions. However, in the case of a heavy-tailed distribution, they demonstrated in simulation study that CDM-PCA tends to perform better than the NR method. There are quite some applications using CDM methods. For instance, a correlation test for high-dimensional data [18], the estimation of $\text{tr}(\Sigma^2)$ in high dimensional setting [20], and a two-sample test under the strongly spiked eigenvalue model ([4] and [8]). Yata and Aoshima [16] also established the asymptotic normality for eigenvalue estimates. However, the asymptotic normality for estimates of PC

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directions is still lacking. In this article, we have extended their work to include the investigation of the asymptotic normality for the leading CDM-based PC directions and to compare it with the asymptotic normality for the classical PCA. We find that both the CDM-PCA and the classical PCA have the same asymptotic normality in HDLSS. However, the former provides consistent and asymptotically normal estimates in a broader HDLSS region.

Our contribution in this article is twofold. First, we derive the asymptotic normality for PC directions in HDLSS setting for both CDM-based and the classical PCA. Second, we provide theoretical comparison and explain why CDM-based method works in a broader HDLSS region than the classical PCA.

The rest of this article is organized as follows. In Section 2, a brief review for CDM-PCA and some existing asymptotic results are given. In Section 3, the asymptotic normality for estimates of PC directions for both CDM-PCA and classical PCA are established. Comparison in asymptotic theory for the CDM-PCA and classical PCA is also provided. In Section 4, numerical examples are presented to demonstrate the behavior of asymptotic normality. In Section 5, further discussions and some concluding remarks are made. A summary list of notation usage and all the proofs are placed in [Appendix](#).

2. A brief view for CDM-PCA

In this section, we will introduce a spiked covariance model and give a brief view for CDM-PCA by Yata and Aoshima [16].

2.1. Probabilistic model

Suppose that $\mathbf{X} = (\mathbf{x}_1, \dots, \mathbf{x}_{2n})$ is a data matrix of size $d \times 2n$ with $\mathbf{x}_i = (x_{1i}, \dots, x_{di})^\top$, $i \in \{1, \dots, 2n\}$, where $\mathbf{x}_i$ s are independent and identically distributed from a d -dimensional multivariate distribution with mean zero and covariance matrix Σ . Let $\mathbf{X} = (\mathbf{X}_{(1)}, \mathbf{X}_{(2)})$, where $\mathbf{X}_{(1)}$ and $\mathbf{X}_{(2)}$, each of size $d \times n$, form a partition for $\mathbf{X}$. Assume that $d \gg n$ in the sense $n = o(d)$. In general, the sample sizes of $\mathbf{X}_{(1)}$ and $\mathbf{X}_{(2)}$ can be different and the total sample size n can be an odd number. For notation simplicity but without loss of generality, assume that the sample sizes in the partition are equal. Denote the eigenvalue decomposition of Σ by $\Sigma = \mathbf{H}\mathbf{A}\mathbf{H}^\top$, where $\mathbf{A}$ is a diagonal matrix with eigenvalues $\lambda_1, \dots, \lambda_d$ and $\mathbf{H} = (\mathbf{h}_1, \dots, \mathbf{h}_d)$ is a matrix of corresponding eigenvectors. Let m be a fixed positive integer. Consider the following spiked covariance model: for $\lambda_1 > \dots > \lambda_m > \lambda_{m+1} \geq \dots \geq \lambda_d$,

$$\lambda_j = a_j d^{\alpha_j} \text{ and } \lambda_j < C \text{ for } j > m, \text{ and } a_j, \alpha_1 \geq \dots \geq \alpha_m, C \text{ are positive constants, } j \in \{1, \dots, m\}. \quad (1)$$

The spiked covariance model in (1) is quite natural. Especially, the model with $\alpha_1 \geq 1/2$ is often observed in microarray data sets. A comprehensive discussion of α_i can be found in Yata and Aoshima [19] and Aoshima et al. [3].

2.2. Cross data matrix method

Define a cross data matrix by $\mathbf{S}_{D(1)} = n^{-1} \mathbf{X}_{(1)}^\top \mathbf{X}_{(2)}$ or $\mathbf{S}_{D(2)} = n^{-1} \mathbf{X}_{(2)}^\top \mathbf{X}_{(1)}$. The procedure of CDM-PCA is described below.

1. Perform the singular value decomposition (SVD) of $\mathbf{S}_{D(1)}$, i.e., $\mathbf{S}_{D(1)} = \sum_{j=1}^d \tilde{\lambda}_j \tilde{\mathbf{u}}_{j(1)} \tilde{\mathbf{u}}_{j(2)}^\top$, where $\tilde{\lambda}_j$ s are singular values and $\tilde{\mathbf{u}}_{j(1)}$ s and $\tilde{\mathbf{u}}_{j(2)}$ s are, respectively, the left and right singular vectors.
2. Calculate $\tilde{\mathbf{h}}_{j(\ell)} = \mathbf{X}_{(\ell)} \tilde{\mathbf{u}}_{j(\ell)} / \sqrt{n \tilde{\lambda}_j}$.
3. Let $\tilde{\lambda}_j$ be the estimate of λ_j and $\tilde{\mathbf{h}}_j / \|\tilde{\mathbf{h}}_j\|$ be the estimate of $\mathbf{h}_j$, $j \in \{1, \dots, n\}$, $\ell \in \{1, 2\}$ where $\tilde{\mathbf{h}}_j = \frac{1}{2} \{\tilde{\mathbf{h}}_{j(1)} + \tilde{\mathbf{h}}_{j(2)}\}$.

Note that Yata and Aoshima [16] used $\frac{1}{2} \{\tilde{\mathbf{h}}_{j(1)} + \text{sign}(\tilde{\mathbf{h}}_{j(1)}^\top \tilde{\mathbf{h}}_{j(2)}) \tilde{\mathbf{h}}_{j(2)}\}$. Since $\tilde{\mathbf{h}}_{j(1)}^\top \tilde{\mathbf{h}}_{j(2)} = \frac{1}{n \tilde{\lambda}_j} \tilde{\mathbf{u}}_{j(1)}^\top \mathbf{X}_{(1)}^\top \mathbf{X}_{(2)} \tilde{\mathbf{u}}_{j(2)} = 1$, we dropped $\text{sign}(\tilde{\mathbf{h}}_{j(1)}^\top \tilde{\mathbf{h}}_{j(2)})$ from the above expression. Let $\mathbf{X} = \mathbf{H}\mathbf{A}^{1/2}\mathbf{Z}$ such that $\mathbf{Z} = (\mathbf{z}_1, \dots, \mathbf{z}_d)^\top$ is a $d \times 2n$ spherical data matrix from a distribution with the identity covariance matrix, where $\mathbf{z}_j^\top = (z_{j1}, \dots, z_{j,2n})$. In other words, $\mathbf{E}\mathbf{z}_j = \mathbf{0}_{2n \times 1}$, $\frac{1}{2n} \mathbf{E}\mathbf{z}_j^\top \mathbf{z}_j = 1$, and $\mathbf{E}\mathbf{z}_j^\top \mathbf{z}_k = 0$ for $j \neq k$. Assume that $\limsup_{1 \leq j \leq d} \mathbf{E}|z_{j1}|^4$ is bounded. The following theorem is due to Yata and Aoshima [16].

Theorem 1 (Theorem 2, Theorem 3 and Remark 6 of Yata and Aoshima [16]). Assume the spiked covariance model (1), where the leading m population eigenvalues are distinct. Then,

$$(a) \text{ for } j \in \{1, \dots, m\}, \sqrt{\frac{2n}{\text{var}(z_{j1}^2)}} \left(\frac{\tilde{\lambda}_j}{\lambda_j} - 1 \right) \Rightarrow N(0, 1),$$

and $\text{sign}(\tilde{\mathbf{h}}_j^\top \tilde{\mathbf{h}}_j) \tilde{\mathbf{h}}_j^\top \tilde{\mathbf{h}}_j - 1 = o_p(1)$ under conditions: $d \rightarrow \infty$ and $n \rightarrow \infty$ if $\alpha_j > 1/2$; $d \rightarrow \infty$ and $d^{2(1-\alpha_j)}/n \rightarrow 0$ if $\alpha_j \in (0, 1/2]$;

$$(b) \text{ for } j \in \{1, \dots, m\}, \text{sign}(\tilde{\mathbf{h}}_j^\top \tilde{\mathbf{h}}_j) \tilde{\mathbf{h}}_j^\top \frac{\tilde{\mathbf{h}}_j}{\|\tilde{\mathbf{h}}_j\|} - 1 = o_p(1)$$

under conditions:

- (i) $d \rightarrow \infty$ and $n \rightarrow \infty$ if $\alpha_j > 1$;
- (ii) $d \rightarrow \infty$ and $d^{1-\alpha_j}/n \rightarrow 0$ if $\alpha_j \in (1/2, 1]$;
- (iii) $d \rightarrow \infty$ and $d^{2(1-\alpha_j)}/n \rightarrow 0$ if $\alpha_j \in (0, 1/2]$.

From [Theorem 1](#), the eigenvalues and directions of CDM-PCA achieve the consistency under conditions (i)–(iii). Note that the conditions given by [Theorem 1](#) are not continuous at $\alpha_j = 1/2$. This phenomenon is mentioned in Remark 4 of Yata and Aoshima [16]. From the proof of Lemma 5 in Yata and Aoshima [16], the convergence rate of the non-spiked matrix needs to be of order $o_p(n^{-1/2})$ to ensure the asymptotic normality of $\tilde{\lambda}_j$. For $\alpha_j > 1/2$, conditions $n \rightarrow \infty$ and $d \rightarrow \infty$ ensure that the non-spiked parts can be ignorable asymptotically. However, for $\alpha_j \leq 1/2$, an extra requirement is needed.

3. Main results: asymptotic normality for PC directions

Recall the spiked covariance model in (1). Following the ideas of Yata and Aoshima [15,16], and Jung and Marron [10], the first m eigenvalues of this model can be further considered as

$$\alpha_1 = \alpha_2 = \cdots = \alpha_{s_1} > \alpha_{s_1+1} = \cdots = \alpha_{s_2} > \cdots > \alpha_{s_{j-1}+1} = \cdots = \alpha_{s_j},$$

where $s_0 = 0$ and $s_1 + \cdots + s_j = m$. That is, eigenvalues with the same growth rate are classified into the same group.

For notation simplicity, denote $\mathbf{H} = (\mathbf{H}_{[1:m]}, \mathbf{H}_{\text{tail}})$, where $\mathbf{H}_{[1:m]} = (\mathbf{h}_1, \dots, \mathbf{h}_m)$ and $\mathbf{H}_{\text{tail}} = (\mathbf{h}_{m+1}, \dots, \mathbf{h}_d)$. In addition, the inner product $\mathbf{v}^\top \tilde{\mathbf{v}}$ of a direction $\mathbf{v}$ and its estimate $\tilde{\mathbf{v}}$ is adjusted by $\text{sign}(\mathbf{v}^\top \tilde{\mathbf{v}}) \mathbf{v}^\top \tilde{\mathbf{v}}$ so that the inner product is nonnegative. Such a convention is adopted throughout this article.

3.1. CDM-PCA

Theorem 2 (CDM-PCA). Assume the spiked covariance model (1), where the leading m population eigenvalues are distinct, and conditions of [Theorem 1\(a\)](#). Consider the j th eigenvector, where $s_{u-1} < j \leq s_u \leq m$. Then

$$\sqrt{2n} \mathbf{H}_{[1:m]}^\top (\tilde{\mathbf{h}}_j - \mathbf{h}_j) \Rightarrow N(\mathbf{0}, \mathbf{M}^{[j]}),$$

where $\mathbf{M}^{[j]}$ is an $m \times m$ matrix with (k, k) th diagonal entries and (k, k') th off-diagonal entries given by

$$\mathbf{M}_{kk}^{[j]} = \begin{cases} \frac{a_k a_j}{(a_k - a_j)^2} E[z_{j1}^2 z_{k1}^2], & k \in (s_{u-1}, s_u], \\ 0, & k \notin (s_{u-1}, s_u], \end{cases} \quad (2)$$

$$\mathbf{M}_{kk'}^{[j]} = \begin{cases} \frac{a_j \sqrt{a_k a_{k'}}}{(a_k - a_j)(a_{k'} - a_j)} E[z_{k1} z_{k'1} z_{j1}^2], & k, k' \in (s_{u-1}, s_u], \\ 0, & k' \notin (s_{u-1}, s_u] \text{ or } k \notin (s_{u-1}, s_u]. \end{cases} \quad (3)$$

If we further assume that $\mathbf{Z}$ comes from a spherical distribution, then $E[z_{k1} z_{k'1} z_{j1}^2] = 0$. Note that $\tilde{\mathbf{h}}_j$ is not a unit vector. More precisely, $\|\tilde{\mathbf{h}}_j\|^2 = 1 + \|\tilde{\mathbf{h}}_{j(1)} - \tilde{\mathbf{h}}_{j(2)}\|^2/4 \geq 1$ since $\tilde{\mathbf{h}}_{j(1)}^\top \tilde{\mathbf{h}}_{j(2)} = 1$. In the following theorem, we will show the asymptotic results for length normalized CDM-PC directions $\tilde{\mathbf{h}}_j / \|\tilde{\mathbf{h}}_j\|$.

Theorem 3. Assume the spiked covariance model (1), where the leading m population eigenvalues are distinct, and conditions of [Theorem 1\(a\)](#). Then,

$$(a) \sqrt{2n} \mathbf{h}_k^\top \left(\frac{\tilde{\mathbf{h}}_j}{\|\tilde{\mathbf{h}}_j\|} - \mathbf{h}_j \right) \Rightarrow N(0, \mathbf{M}_{kk}^{[j]}), \quad k \neq j, \quad k, j \in (s_{u-1}, s_u],$$

where $\mathbf{M}_{kk}^{[j]}$ is given by (2), and

$$\sqrt{2n} \mathbf{h}_k^\top \left(\frac{\tilde{\mathbf{h}}_j}{\|\tilde{\mathbf{h}}_j\|} - \mathbf{h}_j \right) = o_p(1), \quad j \in (s_{u-1}, s_u], \quad k \notin (s_{u-1}, s_u];$$

(b) further assume that $d \rightarrow \infty$ and $d^{2(1-\alpha_j)}/n \rightarrow 0$ if $\alpha_j \in (1/2, 1]$. Then

$$\|\tilde{\mathbf{h}}_j\| = 1 + o_p(n^{-1/2}), \quad \sqrt{2n} \mathbf{h}_j^\top \left(\frac{\tilde{\mathbf{h}}_j}{\|\tilde{\mathbf{h}}_j\|} - \mathbf{h}_j \right) = o_p(1), \quad j \in (s_{u-1}, s_u];$$

$$\tilde{\mathbf{h}}_k^\top \tilde{\mathbf{h}}_j = o_p(n^{-1/4}), \quad k \neq j, \quad k, j \leq m.$$

Moreover,

$$\|\mathbf{H}_{\text{tail}}^\top \tilde{\mathbf{h}}_j\| = o_p(n^{-1/4}), \quad \left\| \mathbf{H}_{\text{tail}}^\top \frac{\tilde{\mathbf{h}}_j}{\|\tilde{\mathbf{h}}_j\|} \right\| = o_p(n^{-1/4}).$$

Note that $\{\tilde{\mathbf{h}}_j\}_{j=1}^m$ do not satisfy the orthogonality property, i.e. $\|\tilde{\mathbf{h}}_j\| \neq 1$ and $\tilde{\mathbf{h}}_j^\top \tilde{\mathbf{h}}_k \neq 0$ for $j \neq k$. When $\tilde{\mathbf{h}}_j$ is normalized to unit length by $\tilde{\mathbf{h}}_j / \|\tilde{\mathbf{h}}_j\|$, it can be seen from [Theorem 3\(b\)](#) that the normalized estimate $\tilde{\mathbf{h}}_j / \|\tilde{\mathbf{h}}_j\|$ requires a stronger condition on relation of d and n when $\alpha_j \in (1/2, 1]$ to achieve the consistency.

Remark 1. When the sizes of $\mathbf{X}_{(1)}$ and $\mathbf{X}_{(2)}$ are different, say, $d \times n_1$ and $d \times n_2$, respectively, a cross data matrix is defined as $\mathbf{S}_{D(1)} = (n_1 n_2)^{-1/2} \mathbf{X}_{(1)}^\top \mathbf{X}_{(2)}$. Assume that $0 < \lim_{n_1, n_2 \rightarrow \infty} n_1/n_2 < \infty$. The definition of $\tilde{\mathbf{h}}_{j(\ell)}$ is modified to $\tilde{\mathbf{h}}_{j(\ell)} = \mathbf{X}_{(\ell)} \tilde{\mathbf{u}}_{j(\ell)} / \sqrt{n_\ell \tilde{\lambda}_j}$, $j \in \{1, \dots, n\}$, $\ell \in \{1, 2\}$. The asymptotic normality in Theorems 2 and 3 are still valid by replacing $\sqrt{2n}$ with $2\sqrt{\frac{n_1 n_2}{n_1 + n_2}}$. A balanced partition, i.e., $\lim_{n_1, n_2 \rightarrow \infty} n_1/n_2 = 1/2$, is optimal among all partitions in terms of asymptotic efficiency.

3.2. Classical PCA

Define the sample covariance matrix $\mathbf{S} = (2n)^{-1} \mathbf{X} \mathbf{X}^\top$ and $\mathbf{S}_D = (2n)^{-1} \mathbf{X}^\top \mathbf{X}$. Let $\hat{\lambda}_j$'s, $\hat{\mathbf{h}}_j$'s are, respectively, the eigenvalues and eigenvectors of $\mathbf{S}$. For classical PCA, the eigenvalues and PC directions are obtained from the sample covariance matrix. We first give the asymptotic normality of PC directions of the classical approach. Next, we compare the asymptotic normality of PC directions between the CDM-based method and the classical method.

Theorem 4 (Theorem 3.1 and Theorem 4.1 of Yata and Aoshima [15]). Assume the spiked covariance model (1), where the leading m population eigenvalues are distinct. Then for $j \in \{1, \dots, m\}$,

$$\sqrt{\frac{2n}{\text{var}(z_{j1}^2)}} \left(\frac{\hat{\lambda}_j}{\lambda_j} - 1 \right) \Rightarrow N(0, 1), \quad \mathbf{h}_j^\top \hat{\mathbf{h}}_j - 1 = o_p(1)$$

under conditions:

(i') $d \rightarrow \infty$ and $n \rightarrow \infty$ if $\alpha_j > 1$;

(ii') $d \rightarrow \infty$ and $d^{2(1-\alpha_j)}/n \rightarrow 0$ if $\alpha_j \in (0, 1]$.

Theorem 5 (PCA). Assume the spiked covariance model (1), where the leading m population eigenvalues are distinct, and conditions (i')–(ii') in Theorem 4. Consider the j th eigenvector, where $s_{u-1} < j \leq s_u \leq m$. Then,

(a) $\sqrt{2n} \mathbf{H}_{[1:m]}^\top (\hat{\mathbf{h}}_j - \mathbf{h}_j) \Rightarrow N(\mathbf{0}, \mathbf{M}^{[j]})$,

where $\mathbf{M}^{[j]}$ is given in Theorem 2;

(b) $\|\mathbf{H}_{\text{tail}}^\top (\hat{\mathbf{h}}_j - \mathbf{h}_j)\| = o_p(n^{-1/4})$.

Corollary 6. Assume the spiked covariance model (1), where the leading m population eigenvalues are distinct, and conditions (i')–(ii') in Theorem 4. Then,

(a) $\sqrt{2n} \mathbf{h}_k^\top (\hat{\mathbf{h}}_j - \mathbf{h}_j) \Rightarrow N(0, \mathbf{M}_{kk}^{[j]})$, $k \neq j$, $k, j \in (s_{u-1}, s_u]$,

where $\mathbf{M}_{kk}^{[j]}$ is given by (2);

(b) $\sqrt{2n} \mathbf{h}_j^\top (\hat{\mathbf{h}}_j - \mathbf{h}_j) = o_p(1)$, $j \in (s_{u-1}, s_u]$.

(c) $\sqrt{2n} \mathbf{h}_k^\top (\hat{\mathbf{h}}_j - \mathbf{h}_j) = o_p(1)$, $j \in (s_{u-1}, s_u]$, $k \notin (s_{u-1}, s_u]$.

Remark 2. The asymptotic behavior of PC directions for classical PCA was also investigated by Wang and Fan [14] under different assumptions of the spiked covariance model. In their work, data is assumed to follow a sub-Gaussian distribution, while in our study only finite fourth moment is required. In their work, $d/(n\lambda_j)$ goes to a nonzero constant c_j , while in our setting $d/(n\lambda_j) \rightarrow 0$ under conditions (i')–(ii').

Remark 3. In general, the population mean of $\mathbf{x}_i$ may not be zero. Let

$$\mathbf{S}_{oD(1)} = \mathbf{S}_{oD(2)}^\top = \frac{1}{n} \left(\mathbf{I}_n - \frac{1}{n} \mathbf{1}_n \mathbf{1}_n^\top \right) \mathbf{X}_{(1)}^\top \mathbf{X}_{(2)} \left(\mathbf{I}_n - \frac{1}{n} \mathbf{1}_n \mathbf{1}_n^\top \right),$$

$$\mathbf{S}_{oD} = \frac{1}{2n} \left(\mathbf{I}_{2n} - \frac{1}{2n} \mathbf{1}_{2n} \mathbf{1}_{2n}^\top \right) \mathbf{X}^\top \mathbf{X} \left(\mathbf{I}_{2n} - \frac{1}{2n} \mathbf{1}_{2n} \mathbf{1}_{2n}^\top \right),$$

where $\mathbf{1}_k = (1, 1, \dots, 1)^\top$ is of size k . Following the same arguments of Corollary 2 in [16], Theorems 1–5 and Corollary 6 are still valid by replacing $\{\mathbf{S}_{D(\ell)}, \mathbf{S}_D\}$ with $\{\mathbf{S}_{oD(\ell)}, \mathbf{S}_{oD}\}$.

3.3. Comparison between CDM-PCA and classical PCA

By Theorems 2 and 5, the CDM-PC direction $\tilde{\mathbf{h}}_j$ and the classical PC direction $\hat{\mathbf{h}}_j$, when projected to the spiked subspace, have the same asymptotic normality with covariance $\mathbf{M}^{[j]}$. However, the valid region of this asymptotic normality is different for the two methods. The CDM-PCA has a broader valid region (described by conditions (i)–(iii)) for asymptotic normality than the classical PCA (described by conditions (i')–(ii')). We explain this phenomenon of broader valid region from two perspectives.

3.3.1. Broader consistency region for CDM-PCA

For easy illustration, consider a simple spiked model with $\lambda_1 = d^{\alpha_1}$ and $\lambda_2 = \cdots = \lambda_d = 1$. Let $\mathbf{S}_D = (2n)^{-1} \mathbf{X}^\top \mathbf{X}$. For classical PCA, the largest empirical eigenvalue can be expressed as follows.

$$\hat{\lambda}_1 = \max_{\substack{\mathbf{e}_1, \mathbf{e}_2 \in \mathbb{R}^{2n} \\ \|\mathbf{e}_1\| = \|\mathbf{e}_2\| = 1}} \mathbf{e}_1^\top \mathbf{S}_D \mathbf{e}_2 = \max_{\substack{\mathbf{e}_1, \mathbf{e}_2 \in \mathbb{R}^{2n} \\ \|\mathbf{e}_1\| = \|\mathbf{e}_2\| = 1}} \lambda_1 \left\{ \frac{1}{2n} \mathbf{e}_1^\top \begin{pmatrix} \mathbf{z}_{1(1)} \\ \mathbf{z}_{1(2)} \end{pmatrix} \begin{pmatrix} \mathbf{z}_{1(1)} \\ \mathbf{z}_{1(2)} \end{pmatrix}^\top \mathbf{e}_2 + \mathbf{e}_1^\top \mathbf{R}_n \mathbf{e}_2 \right\},$$

where

$$\mathbf{R}_n = \frac{1}{2n\lambda_1} \sum_{j=2}^d \begin{pmatrix} \mathbf{z}_{j(1)} \\ \mathbf{z}_{j(2)} \end{pmatrix} \begin{pmatrix} \mathbf{z}_{j(1)} \\ \mathbf{z}_{j(2)} \end{pmatrix}^\top.$$

Therefore, we have the following ratio,

$$\frac{\hat{\lambda}_1}{\lambda_1} = \max_{\substack{\mathbf{e}_1, \mathbf{e}_2 \in \mathbb{R}^{2n} \\ \|\mathbf{e}_1\| = \|\mathbf{e}_2\| = 1}} \left\{ (2n)^{-1} \mathbf{e}_1^\top \begin{pmatrix} \mathbf{z}_{1(1)} \\ \mathbf{z}_{1(2)} \end{pmatrix} \begin{pmatrix} \mathbf{z}_{1(1)} \\ \mathbf{z}_{1(2)} \end{pmatrix}^\top \mathbf{e}_2 + \mathbf{e}_1^\top \mathbf{R}_n \mathbf{e}_2 \right\}.$$

For CDM-PCA, the ratio of the largest empirical eigenvalue to the population one can be expressed as follows.

$$\frac{\tilde{\lambda}_1}{\lambda_1} = \lambda_1^{-1} \max_{\substack{\mathbf{e}_1, \mathbf{e}_2 \in \mathbb{R}^n \\ \|\mathbf{e}_1\| = \|\mathbf{e}_2\| = 1}} \mathbf{e}_1^\top \mathbf{S}_{D(1)} \mathbf{e}_2 = \max_{\substack{\mathbf{e}_1, \mathbf{e}_2 \in \mathbb{R}^n \\ \|\mathbf{e}_1\| = \|\mathbf{e}_2\| = 1}} \left\{ n^{-1} \mathbf{e}_1^\top \begin{pmatrix} \mathbf{z}_{1(1)} \\ \mathbf{z}_{1(2)} \end{pmatrix} \begin{pmatrix} \mathbf{z}_{1(1)} \\ \mathbf{z}_{1(2)} \end{pmatrix}^\top \mathbf{e}_2 + \mathbf{e}_1^\top \tilde{\mathbf{R}}_n \mathbf{e}_2 \right\},$$

where

$$\tilde{\mathbf{R}}_n = \frac{1}{n\lambda_1} \sum_{j=2}^d \mathbf{z}_{j(1)} \mathbf{z}_{j(2)}^\top.$$

For $\hat{\lambda}_1$ and $\tilde{\lambda}_1$ to achieve consistency for λ_1 , we need $\|\mathbf{R}_n\|_F = o_p(1)$ and $\|\tilde{\mathbf{R}}_n\|_F = o_p(1)$, respectively, where $\|\cdot\|_F$ denotes the matrix Frobenius norm. Note that

$$\begin{aligned} \mathbf{R}_n &= \frac{1}{2n\lambda_1} \begin{pmatrix} \sum_{j=2}^d \mathbf{z}_{j(1)} \mathbf{z}_{j(1)}^\top & 0 \\ 0 & \sum_{j=2}^d \mathbf{z}_{j(2)} \mathbf{z}_{j(2)}^\top \end{pmatrix} + \frac{1}{2n\lambda_1} \begin{pmatrix} 0 & \sum_{j=2}^d \mathbf{z}_{j(1)} \mathbf{z}_{j(2)}^\top \\ \sum_{\ell=2}^d \mathbf{z}_{j(2)} \mathbf{z}_{j(1)}^\top & 0 \end{pmatrix} \\ &= \mathbf{D}_n + \frac{1}{2n\lambda_1} \begin{pmatrix} 0 & \sum_{j=2}^d \mathbf{z}_{j(1)} \mathbf{z}_{j(2)}^\top \\ \sum_{\ell=2}^d \mathbf{z}_{j(2)} \mathbf{z}_{j(1)}^\top & 0 \end{pmatrix} = \mathbf{D}_n + \frac{1}{2} \begin{pmatrix} \mathbf{0} & \tilde{\mathbf{R}}_n \\ \tilde{\mathbf{R}}_n & \mathbf{0} \end{pmatrix}, \end{aligned} \quad (4)$$

where $\mathbf{D}_n$ is the first term in the expression for $\mathbf{R}_n$. Since

$$\mathbb{E} \left\| (2n\lambda_1)^{-1} \sum_{j=2}^d \mathbf{z}_{j(\ell)} \mathbf{z}_{j(\ell)}^\top \right\|^2 = O_p \left(\frac{d^2}{\lambda_1^2 n} \right), \quad \ell \in \{1, 2\}$$

and

$$\mathbb{E} \left\| (2n\lambda_1)^{-1} \sum_{j=2}^d \mathbf{z}_{j(1)} \mathbf{z}_{j(2)}^\top \right\|^2 = O_p \left(\frac{d}{\lambda_1^2 n} \right),$$

we have the following proposition.

Proposition 7. As $d \rightarrow \infty$ and $n \rightarrow \infty$,

$$\|\mathbf{D}_n\| = O_p(d^{1-\alpha_1}/\sqrt{n}) \quad \text{and} \quad \|\tilde{\mathbf{R}}_n\| = O_p(d^{1/2-\alpha_1}/\sqrt{n}).$$

The residual matrix $\mathbf{R}_n$ in classical PCA involves $\mathbf{D}_n$, which has a larger order of magnitude than $\tilde{\mathbf{R}}_n$ in CDM-PCA. This is the reason that the CDM-PCA can have a broader valid region for consistency and asymptotic normality.

3.3.2. More spiked eigenvalues in CDM-PCA

Recall that $\tilde{\mathbf{h}}_{j(\ell)} = (n\tilde{\lambda}_j)^{-1/2} \mathbf{X}_{(\ell)}^\top \tilde{\mathbf{u}}_{j(\ell)}$ for $\ell = 1, 2$. We have, for $j \in \{1, \dots, m\}$,

$$\tilde{\lambda}_j^2 \tilde{\mathbf{h}}_{j(1)} = \{n^{-2} \mathbf{X}_{(1)} \mathbf{X}_{(1)}^\top \mathbf{X}_{(2)} \mathbf{X}_{(2)}^\top\} \tilde{\mathbf{h}}_{j(1)} \quad \text{and} \quad \tilde{\lambda}_j^2 \tilde{\mathbf{h}}_{j(2)} = \{n^{-2} \mathbf{X}_{(2)} \mathbf{X}_{(2)}^\top \mathbf{X}_{(1)} \mathbf{X}_{(1)}^\top\} \tilde{\mathbf{h}}_{j(2)}. \quad (5)$$

Note that $\mathbb{E}[n^{-2} \mathbf{X}_{(1)} \mathbf{X}_{(1)}^\top \mathbf{X}_{(2)} \mathbf{X}_{(2)}^\top] = \mathbb{E}[n^{-2} \mathbf{X}_{(2)} \mathbf{X}_{(2)}^\top \mathbf{X}_{(1)} \mathbf{X}_{(1)}^\top] = \Sigma^2$, as $\mathbf{X}_{(1)}$ and $\mathbf{X}_{(2)}$ are independent. Both equalities in (5) can be regarded as the sample analog of

$$\lambda_j^2 \mathbf{h}_j = \Sigma^2 \mathbf{h}_j, \quad j \in \{1, \dots, m\}. \quad (6)$$

Table 1

(Degenerate asymptotic normality) Performance comparison of the estimated PC directions for CDM-PCA and PCA methods. Here $\mathbf{h}_j$, $j \in \{1, 2\}$ are the first two population directions; $\tilde{\mathbf{h}}_j$, $\tilde{\mathbf{h}}_j/\|\tilde{\mathbf{h}}_j\|$, and $\hat{\mathbf{h}}_j$, $j = \{1, 2\}$ are the estimated CDM-PCA (non-normalized), CDM-PCA (normalized), and PCA directions. The mean and standard deviation of the first two projected PC directions for Gaussian and Student- t with the sample size $n = 100$ and the covariate dimension $d = 200$, 2000 over 2000 replicate runs.

	CDM-PCA				PCA	
	$\sqrt{2n}(\mathbf{h}_1^\top \tilde{\mathbf{h}}_1 - 1)$	$\sqrt{2n}(\mathbf{h}_2^\top \tilde{\mathbf{h}}_2 - 1)$	$\sqrt{2n}(\mathbf{h}_1^\top \frac{\tilde{\mathbf{h}}_1}{\ \tilde{\mathbf{h}}_1\ } - 1)$	$\sqrt{2n}(\mathbf{h}_2^\top \frac{\tilde{\mathbf{h}}_2}{\ \tilde{\mathbf{h}}_2\ } - 1)$	$\sqrt{2n}(\mathbf{h}_1^\top \hat{\mathbf{h}}_1 - 1)$	$\sqrt{2n}(\mathbf{h}_2^\top \hat{\mathbf{h}}_2 - 1)$
Gaussian						
$d = 200$						
Mean	0.012	0.010	-0.042	-0.102	-0.042	-0.101
Std	0.053	0.054	0.033	0.035	0.033	0.035
$d = 2000$						
Mean	0.008	0.008	-0.065	-0.187	-0.065	-0.185
Std	0.051	0.053	0.038	0.043	0.038	0.042
Student- t						
$d = 200$						
Mean	0.034	0.025	-0.060	-0.099	-0.064	-0.110
Std	0.127	0.112	0.081	0.076	0.095	0.182
$d = 2000$						
Mean	0.040	0.030	-0.069	-0.148	-0.076	-0.197
Std	0.128	0.108	0.067	0.073	0.192	0.548

As for the classical PCA, we have

$$\hat{\lambda}_j \hat{\mathbf{h}}_j = \{(2n)^{-1} \mathbf{X} \mathbf{X}^\top\} \hat{\mathbf{h}}_j, \quad j \in \{1, \dots, m\}. \quad (7)$$

Its corresponding population analog is $\lambda_j \mathbf{h}_j = \Sigma \mathbf{h}_j$. From (5)–(7), we see that the CDM-PCA corresponds to use Σ^2 , while the classical PCA uses Σ . In other words, the CDM method adopts a power-method like strategy with power exponent 2, which leads to more spiked leading eigenvalues.

4. Numerical study

In this section, we conducted a numerical study to demonstrate the asymptotic normality of the estimated CDM-PCA directions in comparison with the classical PCA. Set the sample size $n = 100$, the covariate dimension $d = 200$, 2000, Σ is a $d \times d$ diagonal matrix of the form $\text{diag}(16 d^{2/3}, 4 d^{2/3}, d^{1/3}, .5 d^{1/3}, 1, \dots, 1)$, and the number of simulation runs is 2000. There are four spiked eigenvalues ($s_1 = 2$, $s_2 = 2$ and $m = 4$). Two types of distributions are considered, multivariate Gaussian and multivariate Student- t with 5 degrees of freedom.

Table 1 presents the performance of $\sqrt{2n} \mathbf{h}_j^\top (\tilde{\mathbf{h}}_j - \mathbf{h}_j)$ (non-normalized CDM-PCA) and $\sqrt{2n} \mathbf{h}_j^\top (\tilde{\mathbf{h}}_j / \|\tilde{\mathbf{h}}_j\| - \mathbf{h}_j)$ (normalized CDM-PCA) for $j = 1, 2$, and $\sqrt{2n} \mathbf{h}_j^\top (\hat{\mathbf{h}}_j - \mathbf{h}_j)$ (classical PCA). The non-normalized CDM-PCA outperforms the normalized CDM-PCA, while there is no significant difference between the normalized CDM-PCA and classical PCA. From previous discussions in Section 3, the non-normalized CDM-PCA has a broader consistency region than the normalized CDM-PCA and the classical PCA has the same consistency region as the normalized CDM-PCA. Furthermore, according to Theorems 3 and 5, both $\sqrt{2n} \mathbf{h}_j^\top (\tilde{\mathbf{h}}_j / \|\tilde{\mathbf{h}}_j\| - \mathbf{h}_j)$ and $\sqrt{2n} \mathbf{h}_j^\top (\hat{\mathbf{h}}_j - \mathbf{h}_j)$ are degenerated as $n, d \rightarrow \infty$. For the case $d = 200$, the reported means and standard derivations are indeed close to zero. For the case $d = 2000$, we observe that $\sqrt{2n}(\mathbf{h}_2^\top \frac{\tilde{\mathbf{h}}_2}{\|\tilde{\mathbf{h}}_2\|} - 1)$ and $\sqrt{2n}(\mathbf{h}_2^\top \hat{\mathbf{h}}_2 - 1)$ are somewhat deviating from zero. One possible reason is that d is too large such that conditions could be violated. Table 2 presents the performance of $\sqrt{2n} \mathbf{h}_k^\top \tilde{\mathbf{h}}_j$ (non-normalized CDM-PCA), $\sqrt{2n} \mathbf{h}_k^\top \tilde{\mathbf{h}}_j / \|\tilde{\mathbf{h}}_j\|$ (normalized CDM-PCA), and $\sqrt{2n} \mathbf{h}_k^\top \hat{\mathbf{h}}_j$ (classical PCA). We see that CDM-PCA slightly outperforms the classical PCA.

In Figs. 1–2, we show the normal probability plot of the quantiles of four estimators: $\sqrt{2n} \mathbf{h}_2^\top \tilde{\mathbf{h}}_1 / \|\tilde{\mathbf{h}}_1\|$, $\sqrt{2n} \mathbf{h}_2^\top \hat{\mathbf{h}}_1$, $\sqrt{2n} \mathbf{h}_1^\top \tilde{\mathbf{h}}_2 / \|\tilde{\mathbf{h}}_2\|$, and $\sqrt{2n} \mathbf{h}_1^\top \hat{\mathbf{h}}_2$ versus the theoretical quantile values from the standard normal distribution. The performances of these four estimators from CDM-PCA and PCA are similar. However, there are more points from PCA deviating from the dash line in Student- t case (see Figs. 3–4). It reveals that estimators from CDM-PCA are more stable than those from PCA, especially when data comes from a distribution with heavy tails.

5. Discussions and concluding remarks

In this article, we investigate the asymptotic behavior of CDM-based PCA in HDLSS setting and compare it with the classical PCA. Our results can serve as a necessary complement to the interesting initiate work of Yata and Aoshima [16]. We show that both CDM-PCA and classical PCA have the same asymptotic covariance for their PC directions. However, the asymptotic normality for the CDM-PCA is valid in a broader region. That is, CDM-PCA is able to distinguish the spiked components from the non-spiked components in more general cases. This might partially explain why CDM-PCA has a better empirical performance than the classical PCA.

Table 2

(Asymptotic normality) Performance comparison of the estimated PC directions for CDM-PCA and PCA methods. Here $\mathbf{h}_j$, $j \in \{1, 2\}$ are the first two population directions; $\tilde{\mathbf{h}}_j$, $\tilde{\mathbf{h}}_j/\|\tilde{\mathbf{h}}_j\|$, and $\hat{\mathbf{h}}_j$, $j = \{1, 2\}$ are the estimated CDM-PCA (non-normalized), CDM-PCA (normalized), and PCA directions. The mean and standard deviation of the first two projected PC directions for Gaussian and Student- t with the sample size $n = 100$ and the covariate dimension $d = 200$, 2000 over 2000 replicate runs. Astd is the theoretical value for asymptotic standard deviation given by $\text{Astd} = \sqrt{\frac{a_1 a_2}{(a_1 - a_2)^2} E[z_{11}^2 z_{21}^2]}$.

	CDM-PCA				PCA	
	$\sqrt{2n}\tilde{\mathbf{h}}_2^\top \tilde{\mathbf{h}}_1$	$\sqrt{2n}\tilde{\mathbf{h}}_1^\top \tilde{\mathbf{h}}_2$	$\sqrt{2n}\tilde{\mathbf{h}}_2^\top \frac{\tilde{\mathbf{h}}_1}{\ \tilde{\mathbf{h}}_1\ }$	$\sqrt{2n}\tilde{\mathbf{h}}_1^\top \frac{\tilde{\mathbf{h}}_2}{\ \tilde{\mathbf{h}}_2\ }$	$\sqrt{2n}\hat{\mathbf{h}}_2^\top \hat{\mathbf{h}}_1$	$\sqrt{2n}\hat{\mathbf{h}}_1^\top \hat{\mathbf{h}}_2$
Gaussian						
$d = 200$						
Mean	0.012	0.036	0.012	0.036	−0.018	0.001
Std	0.688	0.684	0.681	0.680	0.675	0.679
$d = 2000$						
Mean	−0.012	−0.006	−0.012	−0.006	0.013	0.003
Std	0.706	0.703	0.692	0.698	0.689	0.697
Astd = 0.666						
Student- t						
$d = 200$						
Mean	0.016	0.005	0.016	0.005	−0.018	−0.041
Std	0.960	0.979	0.947	0.966	0.997	0.999
$d = 2000$						
Mean	−0.009	0.013	−0.008	0.013	0.004	−0.023
Std	0.927	0.941	0.910	0.929	0.979	0.954
Astd = 1.104						

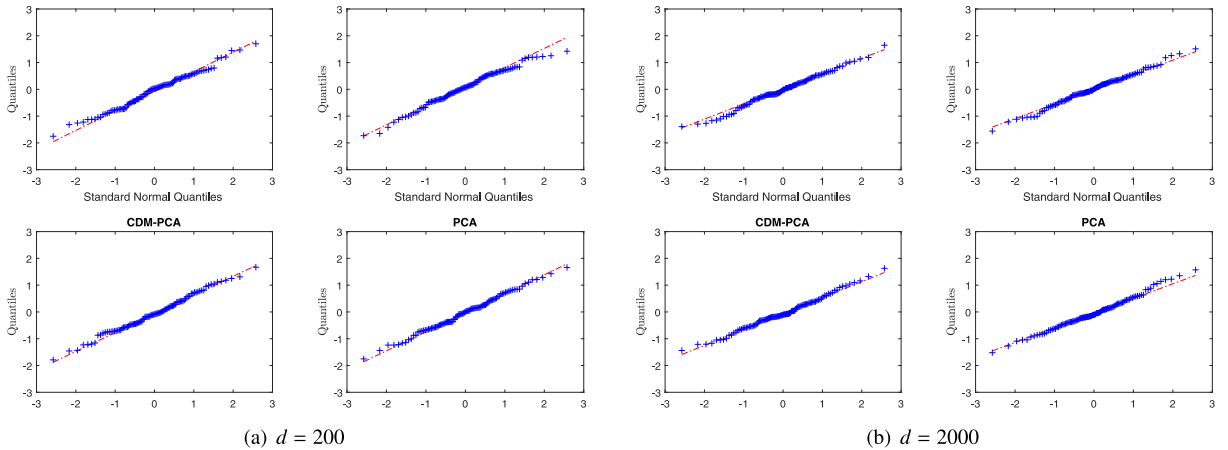


Fig. 1. Here $\mathbf{h}_j$, $j \in \{1, 2\}$ are the first two population directions; $\tilde{\mathbf{h}}_j$ and $\hat{\mathbf{h}}_j$, $j = \{1, 2\}$ are the estimated CDM-PCA (non-normalized) and PCA directions. The normal probability plot for Gaussian case with the sample size $n = 100$ and the covariate dimension $d = 200$, 2000. (upper left) $\sqrt{2n}\tilde{\mathbf{h}}_2^\top \tilde{\mathbf{h}}_1$; (upper right) $\sqrt{2n}\tilde{\mathbf{h}}_1^\top \tilde{\mathbf{h}}_2$; (bottom left) $\sqrt{2n}\tilde{\mathbf{h}}_2^\top \frac{\tilde{\mathbf{h}}_1}{\|\tilde{\mathbf{h}}_1\|}$; (bottom right) $\sqrt{2n}\tilde{\mathbf{h}}_1^\top \frac{\tilde{\mathbf{h}}_2}{\|\tilde{\mathbf{h}}_2\|}$.

To achieve consistency, CDM-PCA requires conditions (i)–(iii) listed in Section 2.2. These conditions imply $d/(\lambda_j) \rightarrow 0$. Wang and Fan [14] have studied the asymptotic behavior of PC directions for classical PCA under the setting $d/(\lambda_j) \rightarrow c_j > 0$. They have discussed bias adjustment to achieve the consistency. It is worth to investigate the asymptotic behavior of CDM-PCA under the setting $d/(\lambda_j) \rightarrow c_j > 0$ in a future study.

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Appendix

A.1. Proofs

The following two lemmas will be used in the proof of Theorems 2 and 5.

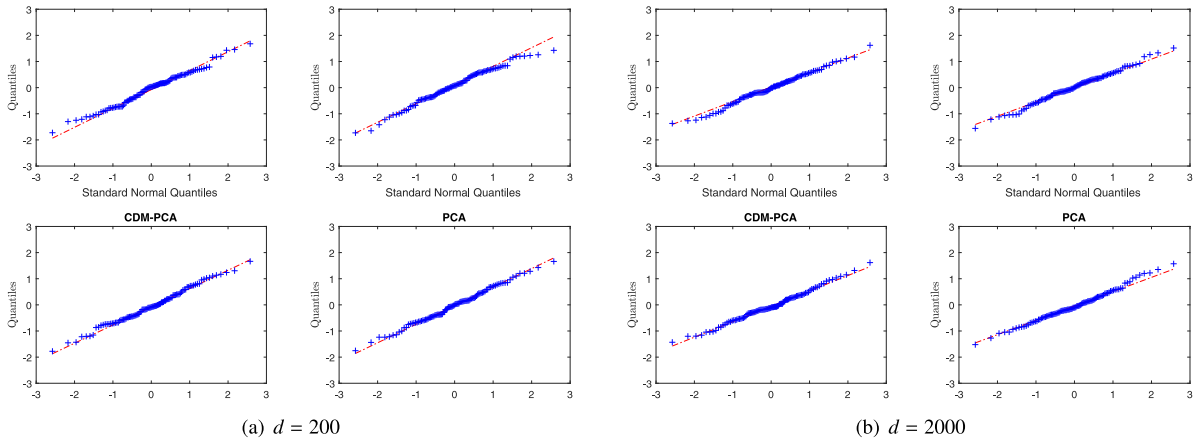


Fig. 2. Here $\mathbf{h}_j$, $j \in \{1, 2\}$ are the first two population directions; $\tilde{\mathbf{h}}_j$ and $\hat{\mathbf{h}}_j$, $j = \{1, 2\}$ are the estimated CDM-PCA (normalized) and PCA directions. The normal probability plot for Gaussian case with the sample size $n = 100$ and the covariate dimension $d = 200, 2000$. (upper left) $\sqrt{2n}\mathbf{h}_2^\top \tilde{\mathbf{h}}_1$; (upper right) $\sqrt{2n}\mathbf{h}_2^\top \hat{\mathbf{h}}_1$; (bottom left) $\sqrt{2n}\mathbf{h}_1^\top \tilde{\mathbf{h}}_2$; (bottom right) $\sqrt{2n}\mathbf{h}_1^\top \hat{\mathbf{h}}_2$.

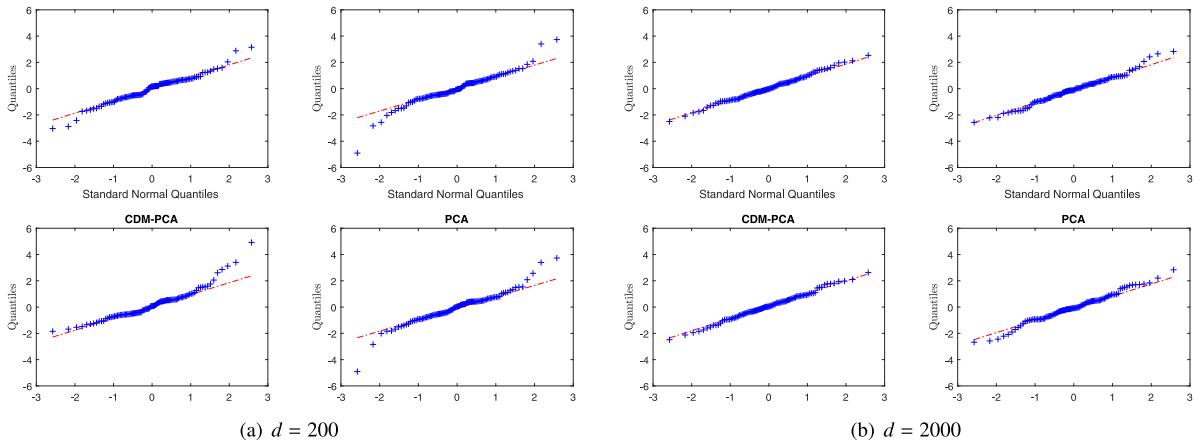


Fig. 3. Here $\mathbf{h}_j$, $j \in \{1, 2\}$ are the first two population directions; $\tilde{\mathbf{h}}_j$ and $\hat{\mathbf{h}}_j$, $j = \{1, 2\}$ are the estimated CDM-PCA (non-normalized) and PCA directions. The normal probability plot for Student- t case with the sample size $n = 100$ and the covariate dimension $d = 200, 2000$. (upper left) $\sqrt{2n}\mathbf{h}_2^\top \tilde{\mathbf{h}}_1$; (upper right) $\sqrt{2n}\mathbf{h}_2^\top \hat{\mathbf{h}}_1$; (bottom left) $\sqrt{2n}\mathbf{h}_1^\top \tilde{\mathbf{h}}_2$; (bottom right) $\sqrt{2n}\mathbf{h}_1^\top \hat{\mathbf{h}}_2$.

Lemma A.1. Assume the spiked covariance model (1), where the leading m population eigenvalues are distinct, and conditions of Theorem 1(a). Then for $j, k \in \{1, \dots, m\}$,

(a) $\sqrt{n}(\mathbf{h}_j^\top \tilde{\mathbf{h}}_j - 1) = o_p(1)$.

(b) $\sqrt{\frac{\tilde{\lambda}_j}{\tilde{\lambda}_k}} \mathbf{h}_k^\top \tilde{\mathbf{h}}_j + \sqrt{\frac{\tilde{\lambda}_k}{\tilde{\lambda}_j}} \mathbf{h}_j^\top \tilde{\mathbf{h}}_k = \sqrt{\frac{1}{\tilde{\lambda}_j \tilde{\lambda}_k}} \mathbf{h}_j^\top \mathbf{S} \mathbf{h}_k + o_p(n^{-1/2})$.

(c) $\frac{\tilde{\lambda}_j}{\tilde{\lambda}_k} \mathbf{h}_k^\top \tilde{\mathbf{h}}_j + \frac{\tilde{\lambda}_k}{\tilde{\lambda}_j} \mathbf{h}_j^\top \tilde{\mathbf{h}}_k = \left(\frac{1}{\tilde{\lambda}_j} + \frac{1}{\tilde{\lambda}_k} \right) \mathbf{h}_j^\top \mathbf{S} \mathbf{h}_k + \left(\sqrt{\frac{\tilde{\lambda}_j}{\tilde{\lambda}_k}} + \sqrt{\frac{\tilde{\lambda}_k}{\tilde{\lambda}_j}} \right) o_p(n^{-1/2})$.

Proof. (a) Based on Yata and Aoshima [16, equation (A6)], we have

$$\left(\frac{\mathbf{z}_{j(\ell)}}{\|\mathbf{z}_{j(\ell)}\|} \right)^\top \tilde{\mathbf{u}}_{j(\ell)} = 1 + o_p(n^{-1/2}), \quad j \in \{1, \dots, m\}, \quad \ell \in \{1, 2\}. \quad (\text{A.1})$$

It implies that there exist an $n \times 1$ normalized vector $\mathbf{y}_j$ and two random variables b_{1n}, b_{2n} such that $\mathbf{y}_j^\top \mathbf{z}_{j(\ell)} = 0$ and

$$\tilde{\mathbf{u}}_{j(\ell)} = b_{1n} \frac{\mathbf{z}_{j(\ell)}}{\|\mathbf{z}_{j(\ell)}\|} + b_{2n} \mathbf{y}_j, \quad j \in \{1, \dots, m\}, \quad (\text{A.2})$$

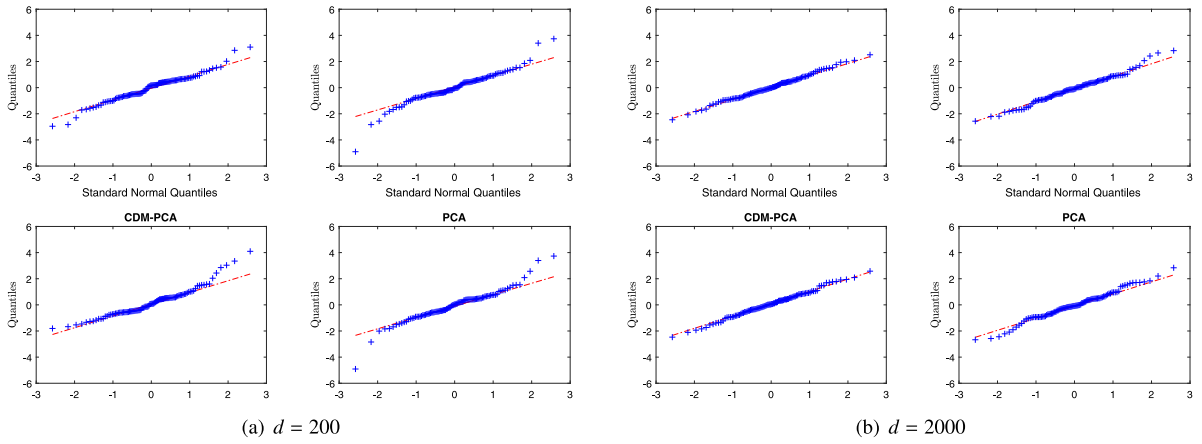


Fig. 4. Here $\mathbf{h}_j$, $j \in \{1, 2\}$ are the first two population directions; $\frac{\tilde{\mathbf{h}}_j}{\|\tilde{\mathbf{h}}_j\|}$ and $\hat{\mathbf{h}}_j$, $j = \{1, 2\}$ are the estimated CDM-PCA (normalized) and PCA directions. The normal probability plot for Student-t case with the sample size $n = 100$ and the covariate dimension $d = 200, 2000$. (upper left) $\sqrt{2n}\mathbf{h}_2^\top \frac{\tilde{\mathbf{h}}_1}{\|\tilde{\mathbf{h}}_1\|}$; (upper right) $\sqrt{2n}\mathbf{h}_2^\top \hat{\mathbf{h}}_1$; (bottom left) $\sqrt{2n}\mathbf{h}_1^\top \frac{\tilde{\mathbf{h}}_2}{\|\tilde{\mathbf{h}}_2\|}$; (bottom right) $\sqrt{2n}\mathbf{h}_1^\top \hat{\mathbf{h}}_2$.

where $b_{1n} = 1 + o_p(n^{-1/2})$ and $b_{2n} = o_p(n^{-1/4})$. Recall that $\mathbf{X}_{(\ell)} = \mathbf{H}\mathbf{\Lambda}^{1/2}\mathbf{Z}_{(\ell)}$ and $\tilde{\mathbf{h}}_{j(\ell)} = \mathbf{X}_{(\ell)}\tilde{\mathbf{u}}_{j(\ell)}/\sqrt{n\tilde{\lambda}_j}$, $\ell \in \{1, 2\}$. We have

$$\mathbf{h}_j^\top \tilde{\mathbf{h}}_{j(\ell)} = \sqrt{\frac{\lambda_j}{n\tilde{\lambda}_j}} \|\mathbf{z}_{j(\ell)}\| \{1 + o_p(n^{-1/2})\}. \quad (\text{A.3})$$

Again, based on Yata and Aoshima [16, equation (A6)],

$$\frac{\tilde{\lambda}_j}{\lambda_j} = \|n^{-1/2}\mathbf{z}_{j(1)}\| \|n^{-1/2}\mathbf{z}_{j(2)}\| + o_p(n^{-1/2}). \quad (\text{A.4})$$

By (A.3) and (A.4),

$$\begin{aligned} \sqrt{n}(\mathbf{h}_j^\top \tilde{\mathbf{h}}_j - 1) &= \sqrt{n} \left(\mathbf{h}_j^\top \left(\frac{\tilde{\mathbf{h}}_{j(1)} + \tilde{\mathbf{h}}_{j(2)}}{2} \right) - 1 \right) \\ &= \sqrt{n} \frac{(\|\mathbf{z}_{j(1)}\|^{1/2} - \|\mathbf{z}_{j(2)}\|^{1/2})^2}{2\|\mathbf{z}_{j(1)}\|^{1/2}\|\mathbf{z}_{j(2)}\|^{1/2}} (1 + o_p(n^{-1/2})) \\ &= \sqrt{n} \frac{\|\mathbf{z}_{j(1)}\|^2 - \|\mathbf{z}_{j(2)}\|^2}{2n} \left(\frac{\|\mathbf{z}_{j(1)}\|^{1/2} - \|\mathbf{z}_{j(2)}\|^{1/2}}{\|\mathbf{z}_{j(1)}\|^{1/2} + \|\mathbf{z}_{j(2)}\|^{1/2}} \right) \left(\frac{n(1 + o_p(n^{-1/2}))}{\|\mathbf{z}_{j(1)}\|^{1/2}\|\mathbf{z}_{j(2)}\|^{1/2}(\|\mathbf{z}_{j(1)}\| + \|\mathbf{z}_{j(2)}\|)} \right). \end{aligned} \quad (\text{A.5})$$

Since $\mathbf{z}_{j1(\ell)}, \dots, \mathbf{z}_{jn(\ell)}$ are independent and identical for given j ,

$$\sqrt{n} \frac{\|\mathbf{z}_{j(1)}\|^2 - \|\mathbf{z}_{j(2)}\|^2}{2n} = O_p(1), \quad \frac{\|\mathbf{z}_{j(1)}\|^{1/2} - \|\mathbf{z}_{j(2)}\|^{1/2}}{\|\mathbf{z}_{j(1)}\|^{1/2} + \|\mathbf{z}_{j(2)}\|^{1/2}} = o_p(1), \quad \frac{\|\mathbf{z}_{j(1)}\|^{1/2}\|\mathbf{z}_{j(2)}\|^{1/2}(\|\mathbf{z}_{j(1)}\| + \|\mathbf{z}_{j(2)}\|)}{n} = O_p(1). \quad (\text{A.6})$$

By (A.5) and (A.6), we have $\sqrt{n}(\mathbf{h}_j^\top \tilde{\mathbf{h}}_j - 1) = o_p(1)$.

(b) Since $\tilde{\mathbf{h}}_{j(1)}^\top \frac{\mathbf{X}_{(2)}\mathbf{X}_{(2)}^\top}{\tilde{\lambda}_j n} \tilde{\mathbf{h}}_{j(1)} = 1$ and $\frac{\mathbf{X}_{(2)}\mathbf{X}_{(2)}^\top}{\tilde{\lambda}_j n} \tilde{\mathbf{h}}_{j(1)} = \tilde{\mathbf{h}}_{j(2)}$,

$$\begin{aligned} (\tilde{\mathbf{h}}_{j(1)} - \mathbf{h}_j)^\top \frac{\mathbf{X}_{(2)}\mathbf{X}_{(2)}^\top}{\tilde{\lambda}_j n} (\tilde{\mathbf{h}}_{j(1)} - \mathbf{h}_j) &= \tilde{\mathbf{h}}_{j(1)}^\top \frac{\mathbf{X}_{(2)}\mathbf{X}_{(2)}^\top}{\tilde{\lambda}_j n} \tilde{\mathbf{h}}_{j(1)} - 2\mathbf{h}_j^\top \frac{\mathbf{X}_{(2)}\mathbf{X}_{(2)}^\top}{\tilde{\lambda}_j n} \tilde{\mathbf{h}}_{j(1)} + \mathbf{h}_j^\top \frac{\mathbf{X}_{(2)}\mathbf{X}_{(2)}^\top}{\tilde{\lambda}_j n} \mathbf{h}_j \\ &= 1 - 2\mathbf{h}_j^\top \tilde{\mathbf{h}}_{j(2)} + \frac{\lambda_j}{\tilde{\lambda}_j} \frac{\mathbf{z}_{j(2)}^\top \mathbf{z}_{j(2)}}{n}. \end{aligned} \quad (\text{A.7})$$

Similarly,

$$(\tilde{\mathbf{h}}_{j(2)} - \mathbf{h}_j)^\top \frac{\mathbf{X}_{(1)}\mathbf{X}_{(1)}^\top}{\tilde{\lambda}_j n} (\tilde{\mathbf{h}}_{j(2)} - \mathbf{h}_j) = 1 - 2\mathbf{h}_j^\top \tilde{\mathbf{h}}_{j(1)} + \frac{\lambda_j}{\tilde{\lambda}_j} \frac{\mathbf{z}_{j(1)}^\top \mathbf{z}_{j(1)}}{n}. \quad (\text{A.8})$$

By (A.7) and (A.8),

$$\begin{aligned}
 & (\tilde{\mathbf{h}}_{j(1)} - \mathbf{h}_j)^\top \frac{\mathbf{X}_{(2)} \mathbf{X}_{(2)}^\top}{\tilde{\lambda}_j n} (\tilde{\mathbf{h}}_{j(1)} - \mathbf{h}_j) + (\tilde{\mathbf{h}}_{j(2)} - \mathbf{h}_j)^\top \frac{\mathbf{X}_{(1)} \mathbf{X}_{(1)}^\top}{\tilde{\lambda}_j n} (\tilde{\mathbf{h}}_{j(2)} - \mathbf{h}_j) \\
 &= 4(1 - \mathbf{h}_j^\top \tilde{\mathbf{h}}_j) + 2 \left(\frac{\lambda_j}{\tilde{\lambda}_j} \frac{\mathbf{z}_j^\top \mathbf{z}_j}{2n} - 1 \right) \stackrel{\text{Lemma A.1(a)}}{=} 2 \left(\frac{\lambda_j}{\tilde{\lambda}_j} \frac{\mathbf{z}_j^\top \mathbf{z}_j}{2n} - 1 \right) + o_p(n^{-1/2}) \\
 &\stackrel{(A.4)}{=} 2 \left(\frac{\|\mathbf{z}_{(1)}\|^2 + \|\mathbf{z}_{(2)}\|^2}{2\|\mathbf{z}_{(1)}\| \|\mathbf{z}_{(2)}\|} - 1 \right) + o_p(n^{-1/2}) = \frac{(\|\mathbf{z}_{(1)}\| - \|\mathbf{z}_{(2)}\|)^2}{\|\mathbf{z}_{(1)}\| \|\mathbf{z}_{(2)}\|} + o_p(n^{-1/2}) = o_p(n^{-1/2}).
 \end{aligned} \tag{A.9}$$

By the same arguments,

$$(\tilde{\mathbf{h}}_{k(1)} - \mathbf{h}_k)^\top \frac{\mathbf{X}_{(2)} \mathbf{X}_{(2)}^\top}{\tilde{\lambda}_k n} (\tilde{\mathbf{h}}_{k(1)} - \mathbf{h}_k) + (\tilde{\mathbf{h}}_{k(2)} - \mathbf{h}_k)^\top \frac{\mathbf{X}_{(1)} \mathbf{X}_{(1)}^\top}{\tilde{\lambda}_k n} (\tilde{\mathbf{h}}_{k(2)} - \mathbf{h}_k) = o_p(n^{-1/2}). \tag{A.10}$$

By (A.9), (A.10), and the Cauchy inequality, we have

$$(\tilde{\mathbf{h}}_{k(1)} - \mathbf{h}_k)^\top \frac{\mathbf{X}_{(2)} \mathbf{X}_{(2)}^\top}{\sqrt{\tilde{\lambda}_j \tilde{\lambda}_k} n} (\tilde{\mathbf{h}}_{j(1)} - \mathbf{h}_j) + (\tilde{\mathbf{h}}_{k(2)} - \mathbf{h}_k)^\top \frac{\mathbf{X}_{(1)} \mathbf{X}_{(1)}^\top}{\sqrt{\tilde{\lambda}_j \tilde{\lambda}_k} n} (\tilde{\mathbf{h}}_{j(2)} - \mathbf{h}_j) = o_p(n^{-1/2}). \tag{A.11}$$

Since $\tilde{\mathbf{h}}_{j(1)}^\top \tilde{\mathbf{h}}_{j(2)} = 1$ and $\tilde{\mathbf{h}}_{j(1)}^\top \tilde{\mathbf{h}}_{k(2)} = 0$ for $j \neq k$,

$$\begin{aligned}
 & (\tilde{\mathbf{h}}_{k(1)} - \mathbf{h}_k)^\top \frac{\mathbf{X}_{(2)} \mathbf{X}_{(2)}^\top}{\sqrt{\tilde{\lambda}_j \tilde{\lambda}_k} n} (\tilde{\mathbf{h}}_{j(1)} - \mathbf{h}_j) + (\tilde{\mathbf{h}}_{k(2)} - \mathbf{h}_k)^\top \frac{\mathbf{X}_{(1)} \mathbf{X}_{(1)}^\top}{\sqrt{\tilde{\lambda}_j \tilde{\lambda}_k} n} (\tilde{\mathbf{h}}_{j(2)} - \mathbf{h}_j) \\
 &= \left(-\sqrt{\frac{\tilde{\lambda}_j}{\tilde{\lambda}_k}} \tilde{\mathbf{h}}_k^\top \tilde{\mathbf{h}}_{j(2)} - \sqrt{\frac{\tilde{\lambda}_k}{\tilde{\lambda}_j}} \mathbf{h}_j^\top \tilde{\mathbf{h}}_{k(2)} \right) + \left(-\sqrt{\frac{\tilde{\lambda}_j}{\tilde{\lambda}_k}} \tilde{\mathbf{h}}_k^\top \tilde{\mathbf{h}}_{j(1)} - \sqrt{\frac{\tilde{\lambda}_k}{\tilde{\lambda}_j}} \mathbf{h}_j^\top \tilde{\mathbf{h}}_{k(1)} \right) + \frac{2}{\sqrt{\tilde{\lambda}_j \tilde{\lambda}_k}} \mathbf{h}_j^\top \mathbf{S} \mathbf{h}_k \\
 &= 2 \left(-\sqrt{\frac{\tilde{\lambda}_j}{\tilde{\lambda}_k}} \tilde{\mathbf{h}}_k^\top \tilde{\mathbf{h}}_j - \sqrt{\frac{\tilde{\lambda}_k}{\tilde{\lambda}_j}} \mathbf{h}_j^\top \tilde{\mathbf{h}}_k + \frac{1}{\sqrt{\tilde{\lambda}_j \tilde{\lambda}_k}} \mathbf{h}_j^\top \mathbf{S} \mathbf{h}_k \right).
 \end{aligned} \tag{A.12}$$

Thus, (b) follows from (A.11) and (A.12).

(c) We will use the same approach as we used for part (b). Define for $\ell \in \{1, 2\}$

$$\Psi_{(\ell)} = \frac{1}{2} \left\{ \frac{\mathbf{X}_{(\ell)} \mathbf{X}_{(\ell)}^\top}{n} \left(\sum_{\ell=1}^m \lambda_\ell^{-1} \mathbf{h}_\ell \mathbf{h}_\ell^\top \right) + \left(\sum_{\ell=1}^m \lambda_\ell^{-1} \mathbf{h}_\ell \mathbf{h}_\ell^\top \right) \frac{\mathbf{X}_{(\ell)} \mathbf{X}_{(\ell)}^\top}{n} \right\}.$$

Note that $\Psi_{(\ell)}$ is a non-negative definite matrix. We have

$$\begin{aligned}
 (\tilde{\mathbf{h}}_{j(1)} - \mathbf{h}_j)^\top \Psi_{(2)} (\tilde{\mathbf{h}}_{j(1)} - \mathbf{h}_j) &= \sum_{\ell=1}^m \left(\tilde{\mathbf{u}}_{j(2)}^\top \frac{\mathbf{z}_{\ell(2)}}{\sqrt{n}} \right) \left(\tilde{\mathbf{u}}_{j(1)}^\top \frac{\mathbf{z}_{\ell(1)}}{\sqrt{n}} \right) + \frac{\mathbf{z}_{j(2)}^\top \mathbf{z}_{j(2)}}{n} - \sqrt{\frac{\tilde{\lambda}_j}{\lambda_j}} \left(\tilde{\mathbf{u}}_{j(2)}^\top \frac{\mathbf{z}_{j(2)}}{\sqrt{n}} \right) \\
 &\quad - \sqrt{\frac{\lambda_j}{\tilde{\lambda}_j}} \sum_{\ell=1}^m \left(\tilde{\mathbf{u}}_{j(1)}^\top \frac{\mathbf{z}_{\ell(1)}}{\sqrt{n}} \right) \left(\frac{\mathbf{z}_{j(2)}^\top \mathbf{z}_{\ell(2)}}{n} \right),
 \end{aligned} \tag{A.13}$$

$$\begin{aligned}
 (\tilde{\mathbf{h}}_{j(2)} - \mathbf{h}_j)^\top \Psi_{(1)} (\tilde{\mathbf{h}}_{j(2)} - \mathbf{h}_j) &= \sum_{\ell=1}^m \left(\tilde{\mathbf{u}}_{j(1)}^\top \frac{\mathbf{z}_{\ell(1)}}{\sqrt{n}} \right) \left(\tilde{\mathbf{u}}_{j(2)}^\top \frac{\mathbf{z}_{\ell(2)}}{\sqrt{n}} \right) + \frac{\mathbf{z}_{j(1)}^\top \mathbf{z}_{j(1)}}{n} - \sqrt{\frac{\tilde{\lambda}_j}{\lambda_j}} \left(\tilde{\mathbf{u}}_{j(1)}^\top \frac{\mathbf{z}_{j(1)}}{\sqrt{n}} \right) \\
 &\quad - \sqrt{\frac{\lambda_j}{\tilde{\lambda}_j}} \sum_{\ell=1}^m \left(\tilde{\mathbf{u}}_{j(2)}^\top \frac{\mathbf{z}_{\ell(2)}}{\sqrt{n}} \right) \left(\frac{\mathbf{z}_{j(1)}^\top \mathbf{z}_{\ell(1)}}{n} \right).
 \end{aligned} \tag{A.14}$$

By (A.2), (A.4), (A.13), and (A.14),

$$\begin{aligned}
 & (\tilde{\mathbf{h}}_{j(1)} - \mathbf{h}_j)^\top \tilde{\Psi}_{(2)}(\tilde{\mathbf{h}}_{j(1)} - \mathbf{h}_j) + (\tilde{\mathbf{h}}_{j(2)} - \mathbf{h}_j)^\top \tilde{\Psi}_{(1)}(\tilde{\mathbf{h}}_{j(2)} - \mathbf{h}_j) \\
 & \stackrel{(A.2)}{=} 2 \left\{ \left(\frac{\|\mathbf{z}_{j(2)}\| \|\mathbf{z}_{j(1)}\|}{\sqrt{n}} \right) + \frac{\mathbf{z}^\top \mathbf{z}}{2n} - \sqrt{\frac{\tilde{\lambda}_j}{\lambda_j}} \frac{\mathbf{z}^\top \mathbf{z}}{2n} - \sqrt{\frac{\lambda_j}{\tilde{\lambda}_j}} \left(\frac{\|\mathbf{z}_{j(2)}\| \|\mathbf{z}_{j(1)}\|}{\sqrt{n}} \right) \right\} + o_p(n^{-1/2}) \\
 & \stackrel{(A.4)}{=} 2 \left\{ \left(\frac{\tilde{\lambda}_j}{\lambda_j} \right) + \frac{\mathbf{z}^\top \mathbf{z}}{2n} - \left(\frac{\tilde{\lambda}_j}{\lambda_j} \right)^{1/2} \frac{\mathbf{z}^\top \mathbf{z}}{2n} - \left(\frac{\tilde{\lambda}_j}{\lambda_j} \right)^{1/2} \right\} + o_p(n^{-1/2}) \\
 & = 2 \left\{ \left(\frac{\tilde{\lambda}_j}{\lambda_j} \right)^{1/2} - \frac{\mathbf{z}^\top \mathbf{z}}{2n} \right\} \left\{ \left(\frac{\tilde{\lambda}_j}{\lambda_j} \right)^{1/2} - 1 \right\} + o_p(n^{-1/2}) = o_p(n^{-1/2}). \tag{A.15}
 \end{aligned}$$

Similarly,

$$(\tilde{\mathbf{h}}_{k(1)} - \mathbf{h}_k)^\top \tilde{\Psi}_{(2)}(\tilde{\mathbf{h}}_{k(1)} - \mathbf{h}_k) + (\tilde{\mathbf{h}}_{k(2)} - \mathbf{h}_k)^\top \tilde{\Psi}_{(1)}(\tilde{\mathbf{h}}_{k(2)} - \mathbf{h}_k) = o_p(n^{-1/2}). \tag{A.16}$$

By (A.15), (A.16) and the Cauchy inequality, we have

$$(\tilde{\mathbf{h}}_{j(1)} - \mathbf{h}_j)^\top \tilde{\Psi}_{(2)}(\tilde{\mathbf{h}}_{k(1)} - \mathbf{h}_k) + (\tilde{\mathbf{h}}_{j(2)} - \mathbf{h}_j)^\top \tilde{\Psi}_{(1)}(\tilde{\mathbf{h}}_{k(2)} - \mathbf{h}_k) = o_p(n^{-1/2}). \tag{A.17}$$

Obviously,

$$\begin{aligned}
 & (\tilde{\mathbf{h}}_{j(1)} - \mathbf{h}_j)^\top \tilde{\Psi}_{(2)}(\tilde{\mathbf{h}}_{k(1)} - \mathbf{h}_k) + (\tilde{\mathbf{h}}_{j(2)} - \mathbf{h}_j)^\top \tilde{\Psi}_{(1)}(\tilde{\mathbf{h}}_{k(2)} - \mathbf{h}_k) \\
 & = \frac{\tilde{\lambda}_j}{\lambda_k} \mathbf{h}_k^\top \tilde{\mathbf{h}}_j + \frac{\tilde{\lambda}_k}{\lambda_j} \mathbf{h}_j^\top \tilde{\mathbf{h}}_k - \left(\frac{1}{\lambda_j} + \frac{1}{\lambda_k} \right) \mathbf{h}_j^\top \mathbf{S} \mathbf{h}_k \\
 & + \left\{ \sqrt{\frac{\tilde{\lambda}_j}{\lambda_k}} \sum_{\ell=1}^m \left(\mathbf{u}_{j(2)}^\top \frac{\mathbf{z}_{\ell(2)}}{\sqrt{n}} \right) \left(\mathbf{u}_{k(1)}^\top \frac{\mathbf{z}_{\ell(1)}}{\sqrt{n}} \right) - \sqrt{\frac{\lambda_j}{\tilde{\lambda}_k}} \sum_{\ell=1}^m \left(\frac{\mathbf{z}_{j(2)}^\top \mathbf{z}_{\ell(2)}}{\sqrt{n}} \right) \left(\mathbf{u}_{k(1)}^\top \frac{\mathbf{z}_{\ell(1)}}{\sqrt{n}} \right) \right\} \\
 & + \left\{ \sqrt{\frac{\tilde{\lambda}_j}{\lambda_k}} \sum_{\ell=1}^m \left(\mathbf{u}_{j(1)}^\top \frac{\mathbf{z}_{\ell(1)}}{\sqrt{n}} \right) \left(\mathbf{u}_{k(2)}^\top \frac{\mathbf{z}_{\ell(2)}}{\sqrt{n}} \right) - \sqrt{\frac{\lambda_j}{\tilde{\lambda}_k}} \sum_{\ell=1}^m \left(\frac{\mathbf{z}_{j(1)}^\top \mathbf{z}_{\ell(1)}}{\sqrt{n}} \right) \left(\mathbf{u}_{k(2)}^\top \frac{\mathbf{z}_{\ell(2)}}{\sqrt{n}} \right) \right\} \\
 & + \left\{ \sqrt{\frac{\tilde{\lambda}_k}{\lambda_j}} \sum_{\ell=1}^m \left(\mathbf{u}_{k(2)}^\top \frac{\mathbf{z}_{\ell(2)}}{\sqrt{n}} \right) \left(\mathbf{u}_{j(1)}^\top \frac{\mathbf{z}_{\ell(1)}}{\sqrt{n}} \right) - \sqrt{\frac{\lambda_k}{\tilde{\lambda}_j}} \sum_{\ell=1}^m \left(\frac{\mathbf{z}_{k(2)}^\top \mathbf{z}_{\ell(2)}}{\sqrt{n}} \right) \left(\mathbf{u}_{j(1)}^\top \frac{\mathbf{z}_{\ell(1)}}{\sqrt{n}} \right) \right\} \\
 & + \left\{ \sqrt{\frac{\tilde{\lambda}_k}{\lambda_j}} \sum_{\ell=1}^m \left(\mathbf{u}_{k(1)}^\top \frac{\mathbf{z}_{\ell(1)}}{\sqrt{n}} \right) \left(\mathbf{u}_{j(2)}^\top \frac{\mathbf{z}_{\ell(2)}}{\sqrt{n}} \right) - \sqrt{\frac{\lambda_k}{\tilde{\lambda}_j}} \sum_{\ell=1}^m \left(\frac{\mathbf{z}_{k(1)}^\top \mathbf{z}_{\ell(1)}}{\sqrt{n}} \right) \left(\mathbf{u}_{j(2)}^\top \frac{\mathbf{z}_{\ell(2)}}{\sqrt{n}} \right) \right\}.
 \end{aligned}$$

Based on (A.2) and the fact $\mathbf{X}_{(1)}$ and $\mathbf{X}_{(2)}$ are independent,

$$\begin{aligned}
 & (\tilde{\mathbf{h}}_{j(1)} - \mathbf{h}_j)^\top \tilde{\Psi}_{(2)}(\tilde{\mathbf{h}}_{k(1)} - \mathbf{h}_k) + (\tilde{\mathbf{h}}_{j(2)} - \mathbf{h}_j)^\top \tilde{\Psi}_{(1)}(\tilde{\mathbf{h}}_{k(2)} - \mathbf{h}_k) \\
 & = \frac{\tilde{\lambda}_j}{\lambda_k} \mathbf{h}_k^\top \tilde{\mathbf{h}}_j + \frac{\tilde{\lambda}_k}{\lambda_j} \mathbf{h}_j^\top \tilde{\mathbf{h}}_k - \left(\frac{1}{\lambda_j} + \frac{1}{\lambda_k} \right) \mathbf{h}_j^\top \mathbf{S} \mathbf{h}_k + \left(\sqrt{\frac{\tilde{\lambda}_j}{\lambda_k}} + \sqrt{\frac{\lambda_j}{\tilde{\lambda}_k}} \right) o_p(n^{-1/2}). \tag{A.18}
 \end{aligned}$$

Thus, (c) follows from (A.17) and (A.18). $\square$

Lemma A.2. Assume the spiked covariance model (1), where the leading m population eigenvalues are distinct, and conditions (i')–(ii') in Theorem 4. Then for $j, k \in \{1, \dots, m\}$,

(a) $\sqrt{n}(\mathbf{h}_j^\top \tilde{\mathbf{h}}_j - 1) = o_p(1)$.

(b) $\mathbf{h}_k^\top \tilde{\mathbf{h}}_j + \mathbf{h}_j^\top \tilde{\mathbf{h}}_k = o_p(n^{-1/2})$.

(c) $\sqrt{\frac{\tilde{\lambda}_j}{\lambda_k}} \mathbf{h}_k^\top \tilde{\mathbf{h}}_j + \sqrt{\frac{\tilde{\lambda}_k}{\lambda_j}} \mathbf{h}_j^\top \tilde{\mathbf{h}}_k = \sqrt{\frac{1}{\tilde{\lambda}_j \tilde{\lambda}_k}} \mathbf{h}_j^\top \mathbf{S} \mathbf{h}_k + o_p(n^{-1/2})$.

Proof. (a) Since $\hat{\mathbf{h}}_j = \mathbf{X} \hat{\mathbf{u}}_j / \sqrt{2n\hat{\lambda}_j}$ and $\mathbf{X} = \mathbf{H} \mathbf{\Lambda}^{1/2} \mathbf{Z}$,

$$\mathbf{h}_j^\top \hat{\mathbf{h}}_j = \sqrt{\frac{\lambda_j}{2n\lambda_j}} \mathbf{z}_j^\top \hat{\mathbf{u}}_j. \tag{A.19}$$

Based on Yata and Aoshima [15, equation (29)], we have for $j \in \{1, \dots, m\}$,

$$\left(\frac{\mathbf{z}_j}{\|\mathbf{z}_j\|} \right)^\top \hat{\mathbf{u}}_j = 1 + o_p(n^{-1/2}). \quad (\text{A.20})$$

It implies that there exist an $2n \times 1$ normalized vector $\mathbf{y}_j$ and two random variables b_{1n}, b_{2n} such that $\mathbf{y}_j^\top \mathbf{z}_j = 0$ and

$$\hat{\mathbf{u}}_j = b_{1n} \frac{\mathbf{z}_j}{\|\mathbf{z}_j\|} + b_{2n} \mathbf{y}_j, \quad \text{given } j \leq m \quad (\text{A.21})$$

with $b_{1n} = 1 + o_p(n^{-1/2})$ and $b_{2n} = o_p(n^{-1/4})$. Together with (A.19),

$$\mathbf{h}_j^\top \hat{\mathbf{h}}_j = \sqrt{\frac{\lambda_j}{2n\lambda_j}} \mathbf{z}_j^\top \mathbf{z}_j + o_p(n^{-1/2}). \quad (\text{A.22})$$

Again, by Yata and Aoshima [15, equation (29)], we have

$$\frac{\hat{\lambda}_j}{\lambda_j} = \frac{\|\mathbf{z}_j\|^2}{2n} + o_p(n^{-1/2}). \quad (\text{A.23})$$

By (A.22) and (A.23), the proof is completed.

(b) Based on the fact $\|\hat{\mathbf{h}}_j\| = 1$ and part (a),

$$\|\hat{\mathbf{h}}_j - \mathbf{h}_j\|^2 = 2(1 - \hat{\mathbf{h}}_j^\top \mathbf{h}_j) = 2\mathbf{h}_j^\top (\mathbf{h}_j - \hat{\mathbf{h}}_j) = o_p(n^{-1/2}). \quad (\text{A.24})$$

Similarly,

$$\|\hat{\mathbf{h}}_k - \mathbf{h}_k\|^2 = o_p(n^{-1/2}). \quad (\text{A.25})$$

From (A.24) and (A.25) and the fact $\hat{\mathbf{h}}_j^\top \hat{\mathbf{h}}_k = 0$, it is implied by the Cauchy inequality that

$$(\hat{\mathbf{h}}_j^\top \mathbf{h}_k + \hat{\mathbf{h}}_k^\top \mathbf{h}_j)^2 = ((\hat{\mathbf{h}}_j - \mathbf{h}_j)^\top (\hat{\mathbf{h}}_k - \mathbf{h}_k))^2 \leq \|\hat{\mathbf{h}}_j - \mathbf{h}_j\|^2 \|\hat{\mathbf{h}}_k - \mathbf{h}_k\|^2 = o_p(n^{-1}).$$

(c) We will use the same approach as we used for part (b). We have

$$\hat{\lambda}_j^{-1} (\hat{\mathbf{h}}_j - \mathbf{h}_j)^\top \mathbf{S} (\hat{\mathbf{h}}_j - \mathbf{h}_j) = 1 - 2\mathbf{h}_j^\top \hat{\mathbf{h}}_j + \frac{\mathbf{h}_j^\top \mathbf{S} \mathbf{h}_j}{\hat{\lambda}_j} \stackrel{\text{Lemma A.2(a), (A.23)}}{=} (2 - 2\mathbf{h}_j^\top \hat{\mathbf{h}}_j) + \left(\frac{\lambda_j}{\hat{\lambda}_j} \frac{\mathbf{z}_j^\top \mathbf{z}_j}{2n} - 1 \right) = o_p(n^{-1/2}). \quad (\text{A.26})$$

Note that $\hat{\mathbf{S}} \hat{\mathbf{h}}_j = \hat{\lambda}_j \hat{\mathbf{h}}_j$, $j \in \{1, \dots, 2n\}$ and $\mathbf{S} = (2n)^{-1} \mathbf{H} \mathbf{A}^{1/2} \mathbf{Z} \mathbf{Z}^\top \mathbf{A}^{1/2} \mathbf{H}^\top$. Similarly,

$$\hat{\lambda}_k^{-1} (\hat{\mathbf{h}}_k - \mathbf{h}_k)^\top \mathbf{S} (\hat{\mathbf{h}}_k - \mathbf{h}_k) = o_p(n^{-1/2}). \quad (\text{A.27})$$

From (A.26), (A.27) and the Cauchy inequality we have

$$\begin{aligned} & \left(\sqrt{\frac{\hat{\lambda}_j}{\hat{\lambda}_k}} \hat{\mathbf{h}}_k^\top \hat{\mathbf{h}}_j + \sqrt{\frac{\hat{\lambda}_k}{\hat{\lambda}_j}} \mathbf{h}_j^\top \hat{\mathbf{h}}_k - \sqrt{\frac{1}{\hat{\lambda}_j \hat{\lambda}_k}} \mathbf{h}_j^\top \mathbf{S} \mathbf{h}_k \right)^2 = \left(\frac{1}{\sqrt{\hat{\lambda}_j \hat{\lambda}_k}} (\hat{\mathbf{h}}_j - \mathbf{h}_j)^\top \mathbf{S} (\hat{\mathbf{h}}_k - \mathbf{h}_k) \right)^2 \\ & \leq \left\{ \frac{(\hat{\mathbf{h}}_j - \mathbf{h}_j)^\top \mathbf{S} (\hat{\mathbf{h}}_j - \mathbf{h}_j)}{\hat{\lambda}_j} \right\} \left\{ \frac{(\hat{\mathbf{h}}_k - \mathbf{h}_k)^\top \mathbf{S} (\hat{\mathbf{h}}_k - \mathbf{h}_k)}{\hat{\lambda}_k} \right\} = o_p(n^{-1}). \quad \square \end{aligned} \quad (\text{A.28})$$

Proof of Theorem 2. We first show the asymptotic normality for $\sqrt{n} \mathbf{h}_k^\top \tilde{\mathbf{h}}_j$. It follows from Lemma A.1(a)–(b) that

$$\left(1 - \frac{\lambda_j}{\lambda_k} \right) \mathbf{h}_k^\top \tilde{\mathbf{h}}_j = \frac{\lambda_j}{\tilde{\lambda}_j \lambda_k} \mathbf{h}_j^\top \mathbf{S} \mathbf{h}_k + \sqrt{\frac{\lambda_j}{\lambda_k}} o_p(n^{-1/2}) = \sqrt{\frac{\lambda_j}{\lambda_k}} \left(\frac{\mathbf{z}_j^\top \mathbf{z}_k}{2n} \right) (1 + o_p(1)).$$

Thus,

$$\mathbf{h}_k^\top \tilde{\mathbf{h}}_j = \frac{\sqrt{\lambda_j \lambda_k}}{\lambda_k - \lambda_j} \left(\frac{1}{2n} \sum_{i=1}^{2n} z_{ji} z_{ki} \right) (1 + o_p(1)) + \frac{\lambda_k}{\lambda_k - \lambda_j} o_p(n^{-1/2}). \quad (\text{A.29})$$

We have

$$\mathbf{H}_{[1:m]}^\top \tilde{\mathbf{h}}_j = \frac{1}{2n} \sum_{i=1}^{2n} \begin{pmatrix} \frac{\sqrt{\lambda_j \lambda_1}}{\lambda_1 - \lambda_j} z_{ji} z_{1i} \\ \vdots \\ \frac{\sqrt{\lambda_j \lambda_m}}{\lambda_m - \lambda_j} z_{ji} z_{mi} \end{pmatrix} (1 + o_p(1)) + o_p(n^{-1/2}) \mathbf{w} \quad (\text{A.30})$$

for some $m \times 1$ unit vector $\mathbf{w}$.

For $j \neq k$ and $j \neq k'$, $j, k, k' \in (s_{u-1}, s_u]$,

$$\text{Avar} \left(\sqrt{2n} \mathbf{h}_k^\top \tilde{\mathbf{h}}_j \right) = \frac{a_k a_j}{(a_k - a_j)^2} E[z_{j1}^2 z_{k1}^2], \quad (\text{A.31})$$

$$\text{Avar} \left(\sqrt{2n} \mathbf{h}_k^\top \tilde{\mathbf{h}}_j, \sqrt{2n} \mathbf{h}_{k'}^\top \tilde{\mathbf{h}}_j \right) = \frac{a_k a_{k'} a_j}{(a_k - a_j)(a_{k'} - a_j)} E[z_{k1} z_{k'1} z_{j1}^2]. \quad (\text{A.32})$$

For $j \in (s_{u-1}, s_u]$ and $k \notin (s_{u-1}, s_u]$, $\sqrt{2n} \mathbf{h}_k^\top \tilde{\mathbf{h}}_j = o_p(1)$. Thus, the proof is completed. $\square$

Proof of Theorem 3. (a) It is a direct result of Theorem 2.

(b) We have

$$\|\tilde{\mathbf{h}}_j\|^2 = \begin{pmatrix} \frac{\tilde{\mathbf{u}}_{j(1)}}{\sqrt{2}} \\ \frac{\tilde{\mathbf{u}}_{j(2)}}{\sqrt{2}} \end{pmatrix}^\top (2n \tilde{\lambda}_j)^{-1} \mathbf{X}^\top \mathbf{X} \begin{pmatrix} \frac{\tilde{\mathbf{u}}_{j(1)}}{\sqrt{2}} \\ \frac{\tilde{\mathbf{u}}_{j(2)}}{\sqrt{2}} \end{pmatrix}. \quad (\text{A.33})$$

Note that $\tilde{\mathbf{u}}_{j(1)}^\top \mathbf{X}_{(1)}^\top \mathbf{X}_{(2)} \tilde{\mathbf{u}}_{j(2)} / (n \tilde{\lambda}_j) = 1$. Based on (A.2) and (A.21), it implies that there exist a $2n \times 1$ normalized vector $\mathbf{y}_j$ and two random variables b_{1n}, b_{2n} such that $\mathbf{y}_j^\top \hat{\mathbf{u}}_j = 0$ and

$$\begin{pmatrix} \frac{\tilde{\mathbf{u}}_{j(1)}}{\sqrt{2}} \\ \frac{\tilde{\mathbf{u}}_{j(2)}}{\sqrt{2}} \end{pmatrix} = b_{1n} \hat{\mathbf{u}}_j + b_{2n} \mathbf{y}_j, \quad \text{given } j \leq m \quad (\text{A.34})$$

with $b_{1n} = 1 + o_p(n^{-1/2})$ and $b_{2n} = o_p(n^{-1/4})$. Therefore,

$$\|\tilde{\mathbf{h}}_j\|^2 \stackrel{(\text{A.33}), (\text{A.34})}{=} \frac{\hat{\lambda}_j}{\tilde{\lambda}_j} \{1 + o_p(n^{-1/2})\} \stackrel{(\text{A.4}), (\text{A.23})}{=} \frac{\|\mathbf{z}_{(1)}\|^2 + \|\mathbf{z}_{(2)}\|^2}{2\|\mathbf{z}_{(1)}\|\|\mathbf{z}_{(2)}\|} \{1 + o_p(n^{-1/2})\} \stackrel{(\text{A.9})}{=} 1 + o_p(n^{-1/2}). \quad (\text{A.35})$$

Similar to (A.34), there exist a $2n \times 1$ unit vector $\mathbf{y}_k$ and two random variables $b'_{1n} = 1 + o_p(n^{-1/2})$ and $b'_{2n} = o_p(n^{-1/4})$ such that $\mathbf{y}_k^\top \hat{\mathbf{u}}_k = 0$ and

$$\begin{pmatrix} \frac{\tilde{\mathbf{u}}_{k(1)}}{\sqrt{2}} \\ \frac{\tilde{\mathbf{u}}_{k(2)}}{\sqrt{2}} \end{pmatrix} = b'_{1n} \hat{\mathbf{u}}_k + b'_{2n} \mathbf{y}_k, \quad \text{given } k \leq m.$$

Then, for $k \neq j$, and $k, j \leq m$,

$$\tilde{\mathbf{h}}_k^\top \tilde{\mathbf{h}}_j = \frac{1}{2n \sqrt{\tilde{\lambda}_k \tilde{\lambda}_j}} \begin{pmatrix} \frac{\tilde{\mathbf{u}}_{k(1)}}{\sqrt{2}} \\ \frac{\tilde{\mathbf{u}}_{k(2)}}{\sqrt{2}} \end{pmatrix}^\top \mathbf{X}^\top \mathbf{X} \begin{pmatrix} \frac{\tilde{\mathbf{u}}_{j(1)}}{\sqrt{2}} \\ \frac{\tilde{\mathbf{u}}_{j(2)}}{\sqrt{2}} \end{pmatrix} = \frac{1}{2n \sqrt{\tilde{\lambda}_k \tilde{\lambda}_j}} (b'_{1n} \hat{\mathbf{u}}_k + b'_{2n} \mathbf{y}_k)^\top \mathbf{X}^\top \mathbf{X} (b_{1n} \hat{\mathbf{u}}_j + b_{2n} \mathbf{y}_j) = o_p(n^{-1/4}).$$

Note that $(2n)^{-1} \mathbf{X}^\top \mathbf{X} \hat{\mathbf{u}}_j = \hat{\lambda}_j \hat{\mathbf{u}}_j$. Moreover, it is implied by Lemma A.1(a) and (A.35) that

$$\mathbf{h}_j^\top \left(\frac{\tilde{\mathbf{h}}_j}{\|\tilde{\mathbf{h}}_j\|} - \mathbf{h}_j \right) = \frac{\mathbf{h}_j^\top (\tilde{\mathbf{h}}_j - \mathbf{h}_j)}{\|\tilde{\mathbf{h}}_j\|} + \frac{1 - \|\tilde{\mathbf{h}}_j\|}{\|\tilde{\mathbf{h}}_j\|} = o_p(n^{-1/2}).$$

As for $\mathbf{H}_{\text{tail}}^\top \tilde{\mathbf{h}}_j$, it is implied by Lemma A.1(a) and (A.35) that

$$\begin{aligned} \|\mathbf{H}_{\text{tail}}^\top \tilde{\mathbf{h}}_j\|^2 &\leq \|\tilde{\mathbf{h}}_j\|^2 - \|\mathbf{h}_j^\top \tilde{\mathbf{h}}_j\|^2 = (\|\tilde{\mathbf{h}}_j\|^2 - 1) + 1 - \|\mathbf{h}_j^\top (\tilde{\mathbf{h}}_j - \mathbf{h}_j) + 1\|^2 \\ &= (\|\tilde{\mathbf{h}}_j\|^2 - 1) - (\mathbf{h}_j^\top (\tilde{\mathbf{h}}_j - \mathbf{h}_j))^2 - 2\mathbf{h}_j^\top (\tilde{\mathbf{h}}_j - \mathbf{h}_j) = o_p(n^{-1/2}). \end{aligned}$$

Since $\|\tilde{\mathbf{h}}_j\| = 1 + o_p(n^{-1/2})$,

$$\left\| \frac{\mathbf{H}_{\text{tail}}^\top \tilde{\mathbf{h}}_j}{\|\tilde{\mathbf{h}}_j\|} \right\|^2 = o_p(n^{-1/2})$$

and the proof is completed. $\square$

Proof of Theorem 5. (a) First, we will show the asymptotic normality of $\mathbf{h}_k^\top \hat{\mathbf{h}}_j$. It follows from Lemma A.2(b) and (c) that

$$\sqrt{\frac{\lambda_k}{\lambda_j}} o_p(n^{-1/2}) + \left(\sqrt{\frac{\hat{\lambda}_j}{\hat{\lambda}_k}} - \sqrt{\frac{\hat{\lambda}_k}{\hat{\lambda}_j}} \right) \mathbf{h}_k^\top \hat{\mathbf{h}}_j = \sqrt{\frac{1}{\hat{\lambda}_j \hat{\lambda}_k}} \mathbf{h}_j^\top \mathbf{S} \mathbf{h}_k + o_p(n^{-1/2}) = \sqrt{\frac{\lambda_j \lambda_k}{\hat{\lambda}_j \hat{\lambda}_k}} \frac{\mathbf{z}_j^\top \mathbf{z}_k}{2n} + o_p(n^{-1/2}).$$

Thus,

$$\mathbf{h}_k^\top \widehat{\mathbf{h}}_j = \frac{\sqrt{\lambda_j \lambda_k}}{\lambda_j - \lambda_k} \left(\frac{1}{2n} \sum_{i=1}^{2n} z_{ji} z_{ki} \right) (1 + o_p(1)) + o_p(n^{-1/2}). \quad (\text{A.36})$$

Note that $\mathbf{S} = (2n)^{-1} \mathbf{X} \mathbf{X}^\top = (2n)^{-1} \mathbf{H} \mathbf{A}^{1/2} \mathbf{Z} \mathbf{Z}^\top \mathbf{A}^{1/2} \mathbf{H}^\top$. We have

$$\mathbf{H}_{[1:m]}^\top \widehat{\mathbf{h}}_j = \frac{1}{2n} \sum_{i=1}^{2n} \begin{pmatrix} \frac{\sqrt{\lambda_j \lambda_1}}{\lambda_j - \lambda_1} z_{ji} z_{1i} \\ \vdots \\ \frac{\sqrt{\lambda_j \lambda_m}}{\lambda_j - \lambda_m} z_{ji} z_{mi} \end{pmatrix} (1 + o_p(1)) + o_p(n^{-1/2}) \mathbf{w} \quad (\text{A.37})$$

for some $m \times 1$ unit vector $\mathbf{w}$.

For $j \neq k$ and $j \neq k', j, k, k' \in (s_{u-1} + 1, s_u]$,

$$\text{Avar} \left(\sqrt{2n} \mathbf{h}_k^\top \widehat{\mathbf{h}}_j \right) = \frac{a_k a_j}{(a_k - a_j)^2} E[z_{j1}^2 z_{k1}^2], \quad (\text{A.38})$$

$$\text{Avar} \left(\sqrt{2n} \mathbf{h}_k^\top \widehat{\mathbf{h}}_j, \sqrt{2n} \mathbf{h}_{k'}^\top \widehat{\mathbf{h}}_j \right) = \frac{a_k a_{k'} a_j}{(a_k - a_j)(a_{k'} - a_j)} E[z_{k1} z_{k'1} z_{j1}^2]. \quad (\text{A.39})$$

When $j \in (s_{u-1} + 1, s_u]$ and $k \notin (s_{u-1} + 1, s_u]$,

$$\sqrt{2n} \mathbf{h}_k^\top (\widehat{\mathbf{h}}_j - \mathbf{h}_j) = o_p(1). \quad (\text{A.40})$$

(b) By Lemma A.2(a) and the fact $\|\widehat{\mathbf{h}}_j\| = 1$,

$$\|\mathbf{H}_{\text{tail}}^\top \widehat{\mathbf{h}}_j\|^2 \leq 1 - \|\mathbf{h}_j^\top \widehat{\mathbf{h}}_j\|^2 = 1 - \|\mathbf{h}_j^\top (\widehat{\mathbf{h}}_j - \mathbf{h}_j) + 1\|^2 = -\{\mathbf{h}_j^\top (\widehat{\mathbf{h}}_j - \mathbf{h}_j)\}^2 - 2\mathbf{h}_j^\top (\widehat{\mathbf{h}}_j - \mathbf{h}_j) = o_p(n^{-1/2}) \quad (\text{A.41})$$

and the theorem is established. $\square$

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